Soroush Dianaty

Physician-Scientist & PhD Student | Biomedical Informatics & Data Science

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Physician and Biomedical Informatics Ph.D. researcher at Arizona State University advancing computational cardiology, health wearables, and digital health systems. Combines clinical training and frontline primary care experience with machine learning, wearable biosignals, and cardiovascular physiological modeling. Research centers on anchoring continuous wearable sensor streams and clinical data in cardiovascular mechanics and physiological rationale, with a focus on cardio-oncology and evidence-grounded health data interoperability. Bridges the gap between physiological modeling and real-world clinical practice.

Education

PhD, Biomedical Informatics and Data Science Aug 2025 – Present
Arizona State University
Doctor of Medicine (M.D.) Sep 2016 – Jun 2024
Tehran Medical Sciences Branch, Islamic Azad University (IAUTMU)

Skills

Machine Learning & AI:

- Deep Learning & Classical ML
- LLM Fine-Tuning (LoRA, PEFT)
- Prompt & Context Engineering
- LLM Application Development

Programming & Data:

- Python
- SQL
- SAS
- Linux & Git

Health Data & Interoperability:

- FHIR
- SNOMED-CT, LOINC, ICD
- EHR Systems (Epic)

Research Methods:

- Regression Modeling
- Survival Analysis
- Cost-effectiveness Analysis

Publications

Peer-reviewed Articles

1. Dianaty S, Gholami F, Gholamrezaie HR, Mirzaei A. "Cost-effectiveness of plasmapheresis and hemoperfusion in COVID-19 survivors: A six-month follow-up analysis after hospital discharge". *Therapeutic Apheresis and Dialysis*, 30(5): 769–781. 2026. doi:10.1111/1744-9987.14235
2. Lee P, Dhadwal AS, Kaiser M, Dianaty S, Lott E, Singh G, Chern D, Walonoski JA, Grando A. "Assessing the Effectiveness and Scalability of Fast Healthcare Interoperability Resource-Based Granular Data Segmentation Technology". *Applied Clinical Informatics*. 2026. doi:10.1055/a-2863-4129
3. Raiszadeh M, Khosronejad A, Dianaty S, Ghorbani Yekta B. "Novel repair of acute Stanford type B aortic dissection using combined endovascular graft and transfemoral replacement of vascular plug". *Catheterization and Cardiovascular Interventions*, 102(7), 1287–1290. 2023. doi:10.1002/ccd.30849
4. Dianaty S, Khodadadi S, Alimoghaddam R, Mirzaei A. "Comparison of outcomes and costs of extracorporeal blood purification therapies in critically ill COVID-19 patients". *Therapeutic Apheresis and Dialysis*, 27(3), 505–516. 2023. doi:10.1111/1744-9987.13948
5. Khoshgoftar Some Saraii Z, Dianaty S, Rouhollah F, Zare N, Ghorbani Yekta B. "Reproductive status of male rat offspring following exposure to methamphetamine during intrauterine life: An experimental study". *International Journal of Reproductive BioMedicine*, 21(2), 175–184. 2023. doi:10.18502/ijrm.v21i2.12809
6. Yaminifar L, Dianaty S, Shahverdi Z, Yaraghi M, Ghorbani Yekta B. "Differential diagnosis of benign ovarian cysts using tumor markers in serum and cyst fluid". *South East Asia Journal of Medical Sciences*, 5(2), 1–7. 2021.
7. Darabi R, Ghoreshi B, Dianaty S, Motevalli MS. "Stress and menstrual disorders among Iranian medical students: A cross-sectional study". *South East Asia Journal of Medical Sciences*, 5(2), 8–15. 2021.

Conference Abstracts & Proceedings

1. Dianaty S, Kaiser M, Murcko A, Grando A. "Early Evidence for Context-Aware Large Language Models (LLMs) in Sensitive Health Data Classification". In *AcademyHealth Annual Research Meeting 2026*, Seattle, WA. 2026 May 30 – Jun 2.
2. Dianaty S, Soumma SB. "Explainable early prediction of acute kidney injury using first 24-hour physiologic and clinical data". In *2nd Arizona Digital Health Symposium*, Phoenix, AZ. 2026 May 5.
3. Dianaty S, Ghorbani Yekta B. "Effects of Descurainia Sophia products on urinary and renal function". In *Congress of Invention, Patenting and Manufacturing of Experimental, Pharmaceutical and Clinical Products*, Tehran, Iran. 2018 Feb 25.

Book Chapters

- Chapters in Practical Guide to Psychiatric Medications (2nd edition). In *Practical Guide to Psychiatric Medications*, 2nd ed., MTP Psychiatry (2022).
  Four chapters: Introduction to Digital Psychiatry; The COVID-19 Pandemic and Mental Health; Apps and Wearables for Mental Health; Future Advances in Mental Health.

Research Experience

Graduate Research Associate Aug 2026 – Present
SHANDHI Lab · Arizona State University · Funded by AHA · PI: Dr. M. M. H. Shandhi | Tempe, AZ

- Anchoring continuous wearable biosignals and multimodal sensor data (PPG, ECG) to physiological rationale and computational cardiovascular modeling.
- Developing translational digital biomarker pipelines from wearable devices and cardiovascular modeling frameworks for cardio-oncology monitoring and risk stratification.

Graduate Research Associate Nov 2024 – Present
Biomedical Knowledge Management & Decision Support (Wang Lab) · Arizona State University · PI: Dr. Dongwen Wang | Tempe, AZ

- Contributing clinical expertise to knowledge representation (Guideline Elements Model / GEM) and RAG pipelines for AI-assisted living clinical practice guidelines, automating evidence extraction from biomedical literature with human-in-the-loop expert validation.
- Developed patient-facing educational content, clinical logic, and backend scoring algorithms for an online contraception decision aid (MyChoicesForAll) for gender-diverse populations (funded by AHRQ).

Graduate Research Associate Oct 2024 – Jun 2026
Substance Use Health Record Sharing (SHARES) · Arizona State University · Funded by NIDA · PI: Dr. Adela Grando | Tempe, AZ

- Developed and validated EHR segmentation rules to support automated substance-use-disorder record sharing under federal privacy frameworks (42 CFR Part 2).
- Applied clinical domain knowledge to improve structured health data interpretability and co-developed context-aware LLM methods for sensitive health text classification.

Student Researcher Aug 2023 – May 2024
Computational Medicine Center · K. N. Toosi University of Technology | Tehran, Iran

- Led interdisciplinary development of neural networks for cardiovascular disease detection from photoplethysmography (PPG) signals.
- Built end-to-end Python pipelines for acquisition, preprocessing, and analysis of EHR-linked biosignal data.

Executive and Research Intern Mar 2022 – Mar 2023
Universal Scientific Education and Research Network (USERN) · Tehran University of Medical Sciences | Tehran, Iran

- Coordinated logistical operations and research-related content development for international scientific congresses, symposiums, and youth researcher initiatives.
- Built Python automation pipelines for certificate generation, reducing processing time by 60%.

Research Assistant May 2018 – Jan 2020
Neuroscience Research Center · Department of Physiology, IAUTMU | Tehran, Iran

- Conducted data entry, visualization, and statistical analysis in IBM SPSS supporting neuropharmacology research.

Teaching Experience

Graduate Teaching Associate Aug 2025 – Present
College of Health Solutions, Arizona State University

- Supported instruction for BMI 507 (Image Processing and Analysis, Fall 2026), BMI 601 (Health Informatics, Spring 2026), and BMI 201 (Introduction to Clinical Informatics, Fall 2025) through hands-on lab facilitation, assignment and discussion design, grading, and student mentoring.

Invited Instructor Sep 2023
Academic Writing & AI Workshop · Student Scientific Associations of TUMS & IUMS

- Designed and delivered a 6-hour interactive online workshop for 50+ medical students and researchers on leveraging generative AI tools in academic literature synthesis, drafting, and scientific writing.

Workshops & Specialized Training

ChatBCT Problem-Based Learning (PBL) Intensive in Digital Health May 11–28, 2026
Trainee / Participant · NIH T32 ChatBCT Program · Arizona State University (PI: Dr. Matthew Buman) · Online

12-hour intensive, case-based training focused on digital health intervention design, behavioral data integration, study strategy, and interdisciplinary problem-solving.

Ethical Consumer Health AI: FUTURE-AI Framework Implementation Spring 2026
Co-Organizer & Facilitator (Lead: Dr. Adela Grando) · Arizona State University

Co-organized interactive workshop implementing the FUTURE-AI international consensus principles (Fairness, Universality, Traceability, Usability, Robustness, Explainability) for consumer health AI applications.

Epic EHR Workshop Mar 27, 2026
Attendee · Instructors: Craig Norquist, MD, FACEP, FAMIA; Timothy Thompson; Micah Oulton; Timothy Raissian, MD · HonorHealth Network Support Services Center, Phoenix, AZ

Intensive clinical informatics training covering EHR workflows, computerized provider order entry (CPOE), physician documentation, and clinical decision support (CDS) integration.

Strategies for FHIR Adoption Feb 6, 2026
Attendee · Instructors: Russell Leftwich, MD, FAMIA (Senior Clinical Advisor, Interoperability, InterSystems) · Health Futures Center (HFC), ASU / Mayo Clinic, Phoenix, AZ

Comprehensive deep dive into HL7 FHIR implementation architectures, US Core profiles, SMART on FHIR app launch frameworks, and health data interoperability.

Presentation Skills for Scientists Jan 16, 2026
Attendee · Instructors: Scott Litin, MD, MACP (Professor of Medicine Emeritus, Mayo Clinic College of Medicine) · Health Futures Center (HFC), ASU / Mayo Clinic, Phoenix, AZ

Advanced strategies for oral medical research communication, medical discourse structuring, and effective scientific data presentation.

Leadership & Professional Experience

Co-Director of Partnerships Aug 2026 – Present
Nucleate Arizona | Phoenix, AZ

- Leading corporate and institutional partnership strategies to bridge academic research with life science venture capitals, biotech firms, and regional innovation hubs.
- Orchestrating sponsor relations, resource allocation, and ecosystem engagement for the annual Nucleate Arizona BioChallenge.

Clinical Experience

Family Physician Oct 2024 – Jul 2025
Qeshlaq Rural Health Center | Abyek, Qazvin, Iran

- Delivered primary and preventive care for a rural population of 7,500+ across 17 villages and 5 satellite health centers.
- Managed chronic disease, prenatal, adolescent, and mental health care within Iran's national EHR system.
- Coordinated care with multidisciplinary teams across distributed rural service sites.

Clinical Rotations Oct 2020 – May 2024
IAUTMU Hospital Network | Tehran, Iran

- Completed 7,000+ hours of clinical rotations across 6 hospitals and 16 specialties.

Awards & Honors

First Place ($2,500), Arizona BioChallenge — Nucleate Arizona
2025-10-01
Graduate Fellowship ($4,845) — Graduate College, Arizona State University
2025-08-01
Active Member Award — Student Research Committee, IAUTMU
2018-10-01

Licenses & Certifications

Inter-Personal Skills in Engineering Project Management | Ira A. Fulton Schools of Engineering, Arizona State University
Jun 2026 – Aug 2026
Fundamentals of Engineering Project Management | Ira A. Fulton Schools of Engineering, Arizona State University
Jun 2026 – Aug 2026
Medical License (No. 222197) | Medical Council of Iran (IRIMC)
Sep 2024
IRB - Biomedical Research | CITI Program Course
Feb 2026
Human Subjects Research | CITI Program Course
Sep 2023
Responsible Conduct of Research & Data Privacy | CITI Program Course
Sep 2023

Professional Activities

Peer Review

- 2027 Amplify Informatics Conference — 6 papers reviewed
- Frontiers in Medicine — 1 manuscript reviewed
- Frontiers in Pharmacology — 1 manuscript reviewed
- Italian Journal of Gynaecology & Obstetrics — 2 manuscripts reviewed

Memberships

- American Medical Informatics Association (AMIA) — Student Member (Dec 2025 – Present)
- Council of Inclusive Excellence (CIE), ASU College of Health Solutions — Board Member (Oct 2025 – Aug 2026)
- USERN Executive Committee — Executive Committee Member (Mar 2022 – Mar 2023)
- Student Research Committee, IAUTMU — Active Member (Feb 2018 – Jun 2024)